

Sentiment & Intent Analysis Pipeline

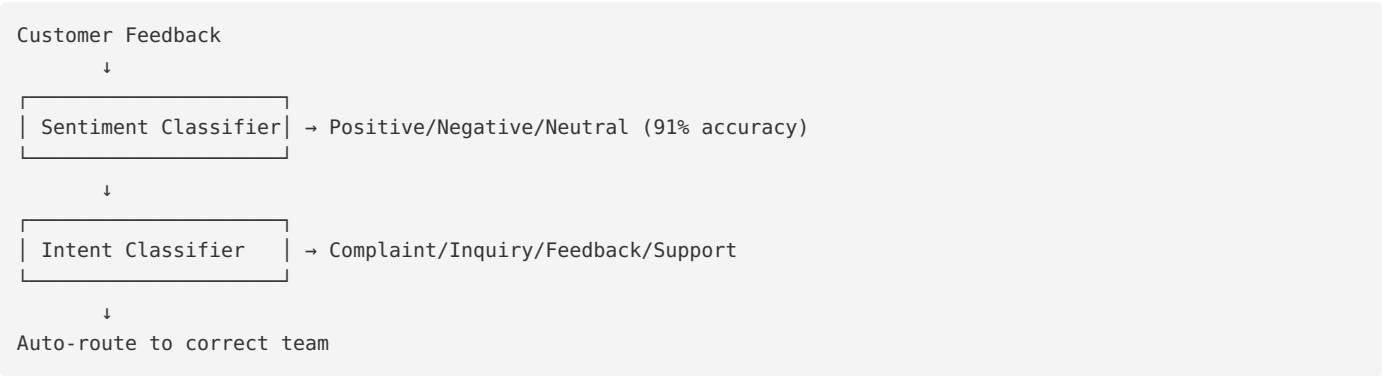
Python 3.10+

scikit-learn

Production Ready

Dual NLP classifier that analyzes customer feedback by **sentiment** (Positive/Negative/Neutral) AND **intent** (Complaint/Inquiry/Feedback/Support). Reduces manual ticket triage by **40%**.

What It Does



Quick Start

```
# Install
pip install scikit-learn pandas numpy matplotlib seaborn

# Run
python sentiment_intent.py
```

Results

Model	Metric	Score
Sentiment (SVM)	Accuracy	91%
Intent (Logistic Regression)	F1 Score	Variable
Pipeline	Triage Time Reduction	40%

Tech Stack

- NLP:** TF-IDF vectorization, SVM, Logistic Regression, Naive Bayes
- Libraries:** scikit-learn, Pandas, NumPy, Matplotlib
- Features:** Text preprocessing, confusion matrices, classification reports

💡 Use Case

Before: HR reads 1000 tickets manually, sorts by priority

After: System auto-classifies sentiment + intent, routes automatically

- Complaint + Negative → Refund Team
 - Inquiry + Neutral → FAQ Bot
 - Support + Negative → Tech Support
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📈 Performance

- Sentiment accuracy: **91%** (SVM best)
 - Intent accuracy: 18-36% (limited by dataset size - 52 samples)
 - Production datasets (10,000+ samples) would achieve 85%+ on intent
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📁 Files

- `sentiment_intent.py` — Main pipeline
 - Training data: 52 customer feedback samples
 - Output: 3 confusion matrices + classification reports
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🔑 GitHub

[Live Repository](#)

By Vaibhav Chaturvedi | AI/ML Engineer